Lecture 13 2025-10-14

$$\hat{p}_n = p_n - \frac{(\Delta p_n)^2}{\Delta^2 p_n} \qquad \Delta p_n = p_{n+1} - p_n$$

$$\hat{\varepsilon}_n = \varepsilon_n - \frac{(\Delta \varepsilon_n)^2}{\Delta^2 \varepsilon_n} = (*)$$

$$= \frac{\varepsilon_n \varepsilon_{n+2} - \varepsilon_{n+1}^2}{\varepsilon_{n+2} - 2\varepsilon_{n+1} + \varepsilon_n}$$

$\nearrow$ 2

$$(*) \overset{(\Delta)}{=} \varepsilon_n - \frac{(\varepsilon_{n+1} - \varepsilon_n)^2}{\varepsilon_{n+2} - 2\varepsilon_{n+1} + \varepsilon_n}$$

$$= \frac{\varepsilon_n(\varepsilon_{n+2} - 2\varepsilon_{n+1} + \varepsilon_n) - (\varepsilon_{n+1}^2 - 2\varepsilon_n\varepsilon_{n+1} + \varepsilon_n^2)}{\varepsilon_{n+2} - 2\varepsilon_{n+1} + \varepsilon_n}$$

$$= \frac{\varepsilon_n \varepsilon_{n+2} - 2\varepsilon_n\varepsilon_{n+1} + \varepsilon_n^2 - \varepsilon_{n+1}^2 + 2\varepsilon_n\varepsilon_{n+1} - \varepsilon_n^2}{\varepsilon_{n+2} - 2\varepsilon_{n+1} + \varepsilon_n}$$

$$= \frac{\varepsilon_n \varepsilon_{n+2} - \varepsilon_{n+1}^2}{\varepsilon_{n+2} - 2\varepsilon_{n+1} + \varepsilon_n}$$

$\varepsilon_n = p_n - p \qquad \hat{\varepsilon}_n = \hat{p}_n - p$

$\Delta\varepsilon_n = p_{n+1} - p_n = \Delta p_n$

$\Delta^2 \varepsilon_n = \Delta\varepsilon_{n+1} - \Delta\varepsilon_n$

$= \varepsilon_{n+2} - \varepsilon_{n+1} - (\varepsilon_{n+1} - \varepsilon_n)$

$= \varepsilon_{n+2} - 2\varepsilon_{n+1} + \varepsilon_n = (\Delta)$

Steffensen's Method: To accelerate convergence of fixed point iteration using a variation of Aitken's Δ^2-method

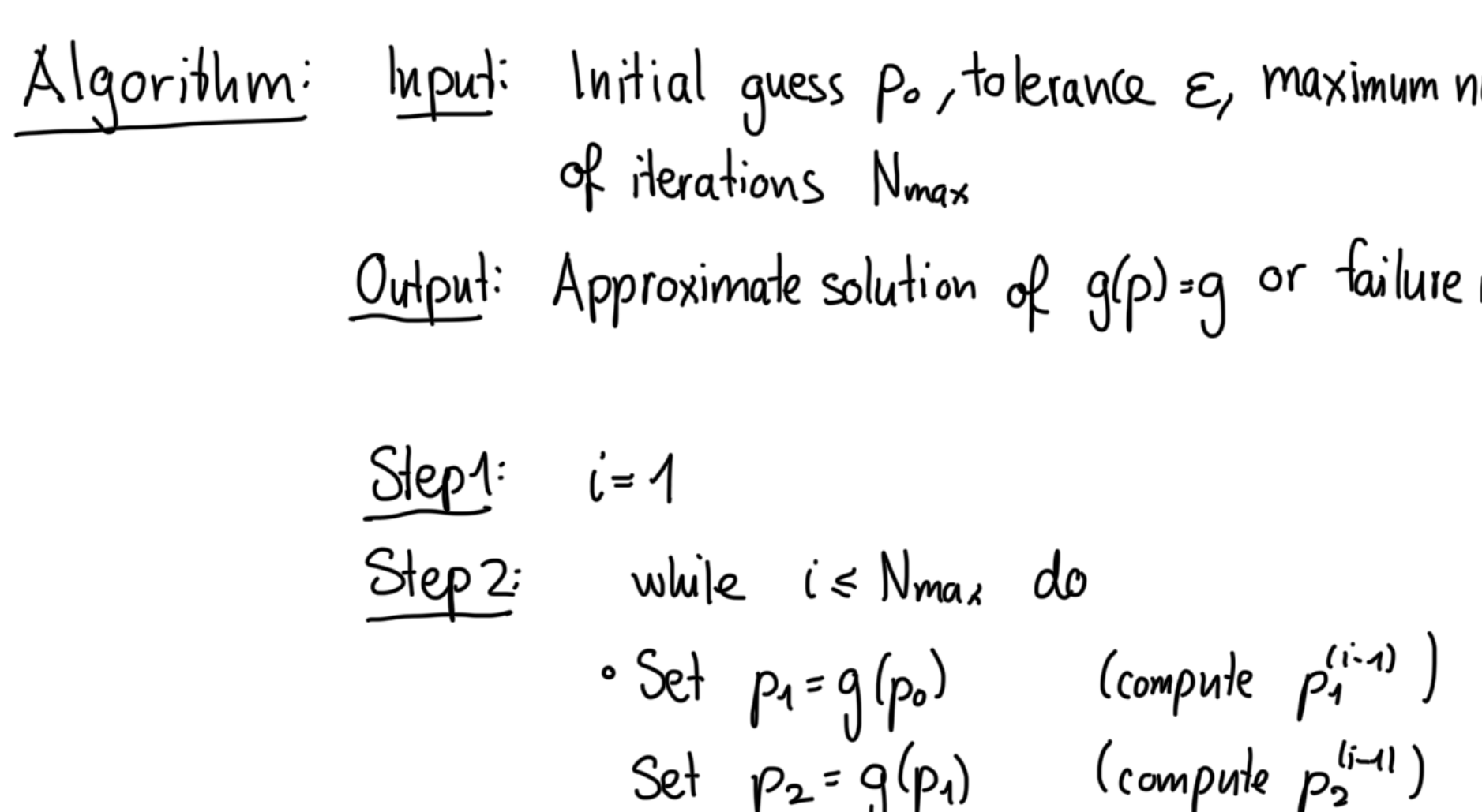
Aitken: generate $p_0, p_1 = g(p_0), p_2 = g(p_1)$

$$\Rightarrow \hat{p}_0 = p_0 - \frac{(\Delta p_0)^2}{\Delta^2 p_0} \qquad p_3 = g(p_2)$$

$$\hat{p}_1 = p_1 - \frac{(\Delta p_1)^2}{} \qquad p_4 = g(p_3)$$

$\Delta^2 p_1$ | 0

Steffensen's Method constructs the same first four terms p_0, p_1, p_2 and $\hat{p}_0$. Then assumes $\hat{p}_0$ is a better approximation than p_2 and applies fixed point iteration to $\hat{p}_0$ instead of p_2.



Algorithm: **Input:** Initial guess p_0, tolerance ε, maximum number of iterations N_{max}

Output: Approximate solution of $g(p) = g$ or failure message

Step 1: $i = 1$

Step 2: while $i \leq N_{max}$ do

- Set $p_1 = g(p_0)$ (compute $p_1^{(i-1)}$)
  Set $p_2 = g(p_1)$ (compute $p_2^{(i-1)}$)
  $$p = p_0 - \frac{(p_1 - p_0)^2}{p_2 - 2p_1 + p_0}$$
- If $|p - p_0| < \varepsilon$
  return p (success!)

- Set $i = i+1$
- Set $p_0 = p$ (update p_0)

Step3: Output "The method failed after N_{max} iterations"

Example: $g(x) = \left(\frac{10}{x+4}\right)^{1/2}$

Theorem 2.15 If $g(p)=p$ and $g'(p) \neq 1$ and $|g''(x)| \leq M$ near p, then $\exists\ \delta > 0$ s.t. Steffensen's Method converges quadratically to p for any $p_0 \in [p-\delta, p+\delta]$.

Proof: A loop of Steffensen's Method can be written as

$$p_0^{(n+1)} = h(p_0^{(n)}) \quad \text{where}$$

$$h(x) = x - \frac{(g(x)-x)^2}{g(g(x)) - 2g(x) + x}$$

Idea: We want to show that $h(p) = p$ and $h'(p) = 0$ provided $g(p)=p$ and $g'(p) \neq 1$.
Then we can use Thm 2.8/2.9 which implies that if $h'(p)=0$, we have quadratic convergence.

Facts: (Taylor expansion) (as $x \to p$)

① $f(x) = O(x-p) \iff f(x^2) = O((x-p)^2)$
② If $f(x) = A(x-p) + O((x-p)^2))$ then $f(x) = O(x-p)$
③ If $f(x) = A + O(x-p)$ & $g(x) = B + O(x-p)$

then $f(x)g(x) = AB + O(x-p)$

④ If $f(x) = A + O(x-p)$ then $\frac{1}{f(x)} = \frac{1}{A} + O(x-p)$ $\quad O(x-p) = K|x-p|$ if $|x-p|$ is small enough

Why? $\left|\frac{1}{A+O(x-p)} - \frac{1}{A}\right| = \frac{1}{A}\left|\frac{\frac{O(x-p)}{A}}{1+O(x-p)}\right| \le \frac{2K}{A^2}|x-p|$

$O(x-p) = a(x)(x-p)$ for some function $a(x)$

and $|a(x)| \le K$ for x small enough

$$\left|\frac{1}{A+a(x)(x-p)} - \frac{1}{A}\right| = \left|\frac{a(x)(x-p)}{A(A+a(x)(x-p)}\right| = \frac{1}{A^2}\left|\frac{a(x)(x-p)}{1+\frac{a(x)}{A}(x-p)}\right|$$

if $|a(x)| \le K$ then $\left|\frac{a(x)(x-p)}{A}\right| \le \frac{1}{2}$ for $|x-p| \le \frac{A}{2K}$

$$= \frac{1}{A}\left|\frac{\frac{a(x)}{A}(x-p)}{1+\frac{a(x)}{A}(x-p)}\right| \le \frac{1}{A}\left|\frac{\frac{K|x-p|}{A}}{\frac{1}{2}}\right| = \frac{2K}{A^2}|x-p| = O(x-p)$$

Now we can write using Taylor's theorem:

(□) $g(x) = p + (x-p)g'(p) + O((x-p)^2)$ $\quad (g(p)=p)$

(○) $g(g(x)) = p + (g(x)-p)\,g'(p) + O((g(x)-p)^2)$ $\quad (g(g(p))=g(p)=p)$ (□)

$= p + (x-p)(g'(p))^2 + O((x-p)^2)$

Numerator: $(g(x)-x)^2 = ((p-x) + (x-p)g'(p) + O((x-p)^2))^2$

$= ((x-p)(g'(p)-1) + O((x-p)^2))^2$

$= ((x-p)(g'(p)-1)(1+O(x-p)))^2$

$= (x-p)^2(g'(p)-1)^2(1+O(x-p))$

Denominator: $g(g(x)) - 2g(x) + x \overset{(□)\&(○)}{=} p + (x-p)g'(p)^2 - 2p - 2(x-p)g'(p) + x + O((x-p)^2)$

$$= (x-p)\left(1-2g'(p)+(g'(p))^2\right)+\mathcal{O}((x-p)^2)$$

$g'(p)\neq 1$

$$= (x-p)\left(1-2g'(p)+(g'(p))^2\right)(1+\mathcal{O}(x-p))$$

$$\Rightarrow \quad \frac{(g(x)-x)^2}{g(g(x))-2g(x)+x} = \frac{(x-p)^2(g'(p)-1)^2\,(1+\mathcal{O}(x-p))}{(x-p)(1-g'(p))^2(1+\mathcal{O}(x-p))}$$

$$= (x-p)\underbrace{\frac{(1+\mathcal{O}(x-p))}{(1+\mathcal{O}(x-p))}}_{\text{bounded for } x-p \text{ small enough}} \overset{④}{=} (x-p)(1+\mathcal{O}(x-p))$$

$$= (x-p)+\mathcal{O}((x-p)^2) \qquad (6)$$

$$h(x) \underset{(6)}{=} x-\left((x-p)+\mathcal{O}((x-p)^2)\right) = p+\mathcal{O}((x-p)^2) \quad = (♡)$$

$$\Rightarrow \lim_{x\to p} h(x) = p \qquad \text{So} \quad h(p)=p$$

$$h'(p) = \lim_{x\to p}\frac{h(x)-h(p)}{x-p} = \lim_{x\to p}\frac{h(x)-p}{x-p} \overset{(♡)}{=} \lim_{x\to p}\mathcal{O}(x-p) = 0$$

∎